

Fernando Zhapa-Camacho

PHD STUDENT · COMPUTER SCIENCE

King Abdullah University of Science and Technology, 4700 KAUST, 23955 Thuwal, Saudi Arabia

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Education

King Abdullah University of Science and Technology

PHD COMPUTER SCIENCE

Thuwal, Saudi Arabia

2022 - present

- Advisor: Robert Hoehndorf

King Abdullah University of Science and Technology

MS COMPUTER SCIENCE

Thuwal, Saudi Arabia

2020 - 2022

- Thesis: Embedding Ontologies Using Category Theory Semantics
- Advisor: Robert Hoehndorf

Yachay Tech University

BE INFORMATION TECHNOLOGY

San Miguel de Urcuquí, Ecuador

2014 - 2020

- Cum laude
- Thesis: Development of a Tropical Algebraic Geometry package in the Haskell programming language.
- Advisor: Francesc Antón Castro

Publications

PREPRINTS

Fernando Zhapa-Camacho, Robert Hoehndorf. 2026. INDIGENA: inductive prediction of disease-gene associations using phenotype ontologies. Preprint.

SELECTED

Fernando Zhapa-Camacho, Robert Hoehndorf. 2026. Fully Geometric Multi-Hop Reasoning on Knowledge Graphs with Transitive Relations. ESWC 2026.

Fernando Zhapa-Camacho, Olga Mashkova, Robert Hoehndorf, Maxat Kulmanov. 2025. LLM Agent Based Protein Function Prediction. PSB 2026.

Fernando Zhapa-Camacho, Robert Hoehndorf. 2025. Lattice-Based ALC Ontology Embeddings With Saturation (Extended Version). Neurosymbolic Artificial Intelligence.

Robert Hoehndorf, Catia Pesquita, **Fernando Zhapa-Camacho**. 2025. Neuro-Symbolic AI in Life Sciences. Frontiers in Artificial Intelligence and Applications.

Jiaoyan Chen, Olga Mashkova, **Fernando Zhapa-Camacho**, Robert Hoehndorf, Yuan He, Ian Horrocks. 2025. Ontology Embedding: A Survey of Methods, Applications and Resources. IEEE TKDE.

Fernando Zhapa-Camacho, Robert Hoehndorf. 2024. Lattice-preserving ALC ontology embeddings. NeSy 2024.

Fernando Zhapa-Camacho, Zhenwei Tang, Maxat Kulmanov, Robert Hoehndorf. 2024. Predicting protein functions using positive-unlabeled ranking with ontology-based priors. ISMB 2024.

Fernando Zhapa-Camacho, Robert Hoehndorf. 2023. From axioms over graphs to vectors, and back again: evaluating the properties of graph-based ontology embeddings. NeSy 2023.

Fernando Zhapa-Camacho, Maxat Kulmanov, Robert Hoehndorf. 2023. mOWL: Python library for machine learning with biomedical ontologies. Bioinformatics, Volume 39, Issue 1.

ADDITIONAL

Olga Mashkova, **Fernando Zhapa-Camacho**, Robert Hoehndorf. 2024. Enhancing Geometric Ontology Embeddings for EL++ with Negative Sampling and Deductive Closure Filtering. NeSy 2024.

Azza Althagafi, **Fernando Zhapa-Camacho**, Robert Hoehndorf. 2024. Prioritizing genomic variants through neuro-symbolic, knowledge-enhanced learning. *Bioinformatics*, Volume 40, Issue 5.

Olga Mashkova, **Fernando Zhapa-Camacho**, Robert Hoehndorf. 2024. DELE: Deductive EL++ Embeddings for Knowledge Base Completion. Preprint.

Fernando Zhapa-Camacho, Robert Hoehndorf. 2023. Evaluating Different Methods for Semantic Reasoning Over Ontologies. *SemREC 2023*.

Sarah M. Alghamdi, **Fernando Zhapa-Camacho**, Robert Hoehndorf. 2022. A-LION - Alignment Learning through Inconsistency negatives of the aligned Ontologies. *The 17th International Workshop on Ontology Matching*.

Maxat Kulmanov, **Fernando Zhapa-Camacho**, Robert Hoehndorf. 2021. DeepGOWeb: fast and accurate protein function prediction on the (Semantic) Web. *Nucleic Acids Research*.

Joseph R. González, **Fernando Zhapa-Camacho**, Oscar V. Guarnizo, Francisco Ortega-Zamorano. 2018. Successive Adaptive Linear Neural Modeling for Equidistant Real Roots Finding. *ETCM 2018*.

Awards

- 2025 **Dean List of the CEMSE Division**, King Abdullah University of Science and Technology
- 2024 **Dean List of the CEMSE Division**, King Abdullah University of Science and Technology
- 2023 **Travel Fellowship BioHackathon Europe**, ELIXIR Europe
Winner of the Semantic Reasoning Evaluation Challenge, International Semantic Web Conference
- 2022 **Travel Fellowship BioHackathon Europe**, ELIXIR Europe

Presentations

Fernando Zhapa-Camacho. 2024. Spotlight paper presentation: Lattice preserving ALC ontology embeddings. *NeSy 2024*.

Maxat Kulmanov, Robert Hoehndorf, Sarah Alghamdi, Azza Althagafi, Sumyyah Toonsi, **Fernando Zhapa-Camacho**. 2023. Tutorial: Machine Learning with Ontologies. *International Conference on Biomedical Ontology 2023*.

Maxat Kulmanov, **Fernando Zhapa-Camacho**. 2023. Tutorial: mOWL - Machine Learning Library with Ontologies. *BHMENA, 2023*.

Sarah Alghamdi, Robert Hoehndorf, Maxat Kulmanov, Sumyyah Toonsi, **Fernando Zhapa-Camacho**. 2023. Tutorial: Machine Learning with Biomedical Ontologies. *SWAT4HCLS, 2023*.

Fernando Zhapa-Camacho. 2023. Spotlight paper presentation: From axioms over graphs to vectors, and back again: evaluating the properties of graph-based ontology embeddings. *NeSy 2023*.

Robert Hoehndorf, Maxat Kulmanov, Sumyyah Toonsi, **Fernando Zhapa-Camacho**, Sarah Alghamdi. 2022. Tutorial: Machine Learning with Biomedical Ontologies. *SWAT4HCLS, 2022 (Virtual)*.

Fernando Zhapa-Camacho. 2020. Abstract presentation: Purely functional implementation of a tropical geometry system in Haskell. *CASC 2020, (Virtual)*.

Fernando Zhapa-Camacho. 2019. Workshop presentation: Development of a Tropical Algebraic Geometry package in the Haskell programming language. *Queen Mary University of London, 2019*.

Fernando Zhapa-Camacho. 2018. Paper presentation: Successive Adaptive Linear Neural Modeling for Equidistant Real Roots Finding. *ETCM, 2018*.

Teaching Experience

Spring 2025 **CS249 Algorithms in Bioinformatics**, Graduate Teaching Assistant
 Spring 2025 **CS321 Application of AI in Bioinformatics**, Graduate Teaching Assistant
 Fall 2023 **CS220 Data Analytics**, Graduate Teaching Assistant
 Fall 2022 **CS249 Algorithms in Bioinformatics**, Graduate Teaching Assistant
 Fall 2021 **CS220 Data Analytics**, Graduate Teaching Assistant
 Spring 2019 **Functional Programming**, Undergraduate Teaching Assistant
 Fall 2017 **Probability and Statistics**, Undergraduate Teaching Assistant

Software Projects

- **mOWL**: A Python library for machine learning with ontologies. Lead developer.

Technical Skills

- **Programming Languages**: Python, Java, Scala
- **Tools**: Git, W&B
- **Libraries**: PyTorch, OWLAPI

Outreach and Extracurricular Activities

SERVICE AND OUTREACH

2022 - 2024 **Yachay Tech Alumni Association**, Committee Member

PEER REVIEW

Bioinformatics

Journal of Biomedical Semantics

PLOS ONE

International Conference on Neural-Symbolic Learning and Reasoning

European Conference on Artificial Intelligence

AAAI Fall Symposium Series

Neurosymbolic Artificial Intelligence